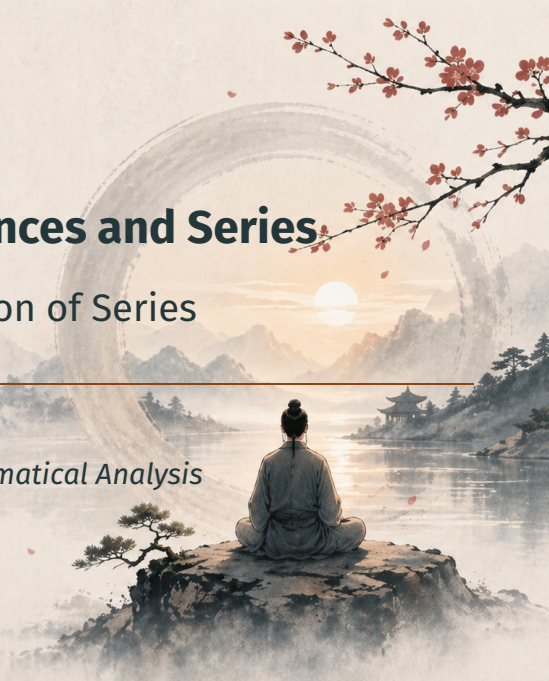


Chapter 3: Numerical Sequences and Series

Section: Addition and Multiplication of Series

Based on Walter Rudin's *Principles of Mathematical Analysis*



Outline

Motivation

Adding Series

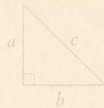
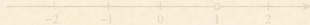
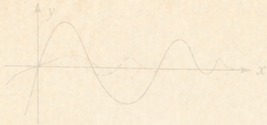
The Cauchy Product

A Product That Diverges

Mertens' Theorem

Abel's Theorem

Summary



$$a^2 + b^2 = c^2$$

Motivation



$$f(x) = \sin x$$



$$\sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{\pi^2}{6}$$



Can We Do Arithmetic with Series?

- For **sequences**, limits respect arithmetic (Theorem 3.3):

$$s_n \rightarrow s, t_n \rightarrow t \implies s_n + t_n \rightarrow s + t, \quad s_n t_n \rightarrow st$$

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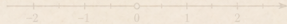
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- For **series**: addition is easy — partial sums simply add
- Multiplication is harder: what should $(\sum a_n) \cdot (\sum b_n)$ even **mean**?
- **Preview**: the **Cauchy product** can diverge (3.49), but **absolute convergence** of one factor saves it (Mertens, 3.50)

$$f(x) = \sin x$$

$$\sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{\pi^2}{6}$$

Adding Series



$$x^2 + y^2 = r^2$$



$$a^2 + b^2 = c^2$$



$$\sum_{n=1}^{\infty} \frac{1}{n^2}$$



Theorem 3.47: Term-by-Term Addition

Theorem 3.47

If $\sum a_n = A$ and $\sum b_n = B$, then

$$\sum (a_n + b_n) = A + B, \quad \text{and} \quad \sum c a_n = cA \text{ for any fixed } c.$$

Key idea: partial sums **add**, and limits respect sums.

Proof of Theorem 3.47

Proof

Let $A_n = \sum_{k=0}^n a_k$ and $B_n = \sum_{k=0}^n b_k$. Then

$$A_n + B_n = \sum_{k=0}^n (a_k + b_k).$$

Since $A_n \rightarrow A$ and $B_n \rightarrow B$, Theorem 3.3 gives

$$\lim_{n \rightarrow \infty} (A_n + B_n) = A + B.$$

For the second claim: $cA_n \rightarrow cA$. ■

From Addition to Multiplication

The situation changes

- Addition: **settled** — convergent series add term by term
- Multiplication: we must first *define* the product of $\sum a_n$ and $\sum b_n$
- Several definitions are possible; we use the “**Cauchy product**”



$$a^2 + b^2 = c^2$$



$$\sum_{n=1}^{\infty} \frac{1}{n^2}$$

The Cauchy Product



Definition 3.48: The Cauchy Product

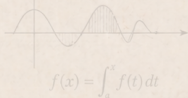
Definition 3.48

Given $\sum a_n$ and $\sum b_n$, put

$$c_n = \sum_{k=0}^n a_k b_{n-k} \quad (n = 0, 1, 2, \dots)$$

and call $\sum c_n$ the *product* of the two given series.

Each c_n collects **all** products $a_i b_j$ with $i + j = n$.



Why This Definition? Power Series

Multiply two power series and collect equal powers of z :

$$\begin{aligned}\sum_{n=0}^{\infty} a_n z^n \cdot \sum_{n=0}^{\infty} b_n z^n &= (a_0 + a_1 z + a_2 z^2 + \cdots)(b_0 + b_1 z + b_2 z^2 + \cdots) \\ &= a_0 b_0 + (a_0 b_1 + a_1 b_0) z + (a_0 b_2 + a_1 b_1 + a_2 b_0) z^2 + \cdots\end{aligned}$$

Why This Definition? Power Series

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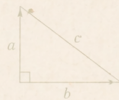
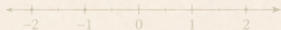
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Setting $z = 1$ yields exactly Definition 3.48.

The Cauchy Product, Visually

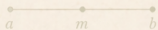
	a_0b_0	a_1b_0	a_2b_0	a_3b_0
c_0	a_0b_1	a_1b_1	a_2b_1	a_3b_1
c_1	a_0b_2	a_1b_2	a_2b_2	a_3b_2
c_2	a_0b_3	a_1b_3	a_2b_3	a_3b_3
c_3				

$$c_n = \sum_{i+j=n} a_i b_j \quad (\text{one anti-diagonal per term})$$



$$a^2 + b^2 = c^2$$

A Product That Diverges



$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$\sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{\pi^2}{6}$$



Example 3.49 — The Question

Example 3.49: Setup

$$\text{Let } A_n = \sum_{k=0}^n a_k, \quad B_n = \sum_{k=0}^n b_k, \quad C_n = \sum_{k=0}^n c_k, \quad \text{with } A_n \rightarrow A, B_n \rightarrow B.$$

- It is **not at all clear** that $C_n \rightarrow AB$, since $C_n \neq A_n B_n$
- The dependence of $\{C_n\}$ on $\{A_n\}, \{B_n\}$ is complicated

Example 3.49 — A Convergent Series

The factor series

$$\sum_{n=0}^{\infty} \frac{(-1)^n}{\sqrt{n+1}} = 1 - \frac{1}{\sqrt{2}} + \frac{1}{\sqrt{3}} - \frac{1}{\sqrt{4}} + \dots$$

converges by the alternating series test (Theorem 3.43).

- Convergence is **not absolute**: $\sum \frac{1}{\sqrt{n+1}}$ diverges
- We form the Cauchy product of this series **with itself**

Example 3.49 — Squaring the Series

The product terms are

$$\sum_{n=0}^{\infty} c_n = 1 - \left(\frac{1}{\sqrt{2}} + \frac{1}{\sqrt{2}} \right) + \left(\frac{1}{\sqrt{3}} + \frac{1}{\sqrt{2}\sqrt{2}} + \frac{1}{\sqrt{3}} \right) - \dots$$

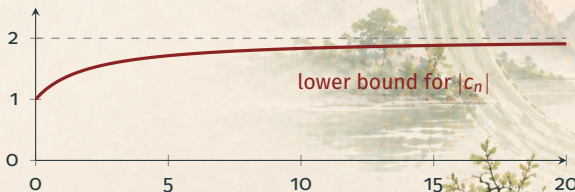
so that

$$c_n = (-1)^n \sum_{k=0}^n \frac{1}{\sqrt{(n-k+1)(k+1)}}$$

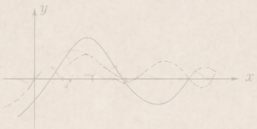
Example 3.49 — The Terms Do Not Vanish

Since $(n - k + 1)(k + 1) = \left(\frac{n}{2} + 1\right)^2 - \left(\frac{n}{2} - k\right)^2 \leq \left(\frac{n}{2} + 1\right)^2$,

$$|c_n| \geq \sum_{k=0}^n \frac{2}{n+2} = \frac{2(n+1)}{n+2} \rightarrow 2$$



$c_n \not\rightarrow 0$, so $\sum c_n$ **diverges** (Theorem 3.23)!

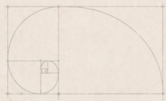


$$e^{i\theta} = \cos \theta + i \sin \theta$$



$$\nabla^2 \phi = 0$$

Mertens' Theorem



$$\sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{\pi^2}{6}$$

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} (a_n \cos nx + b_n \sin nx)$$

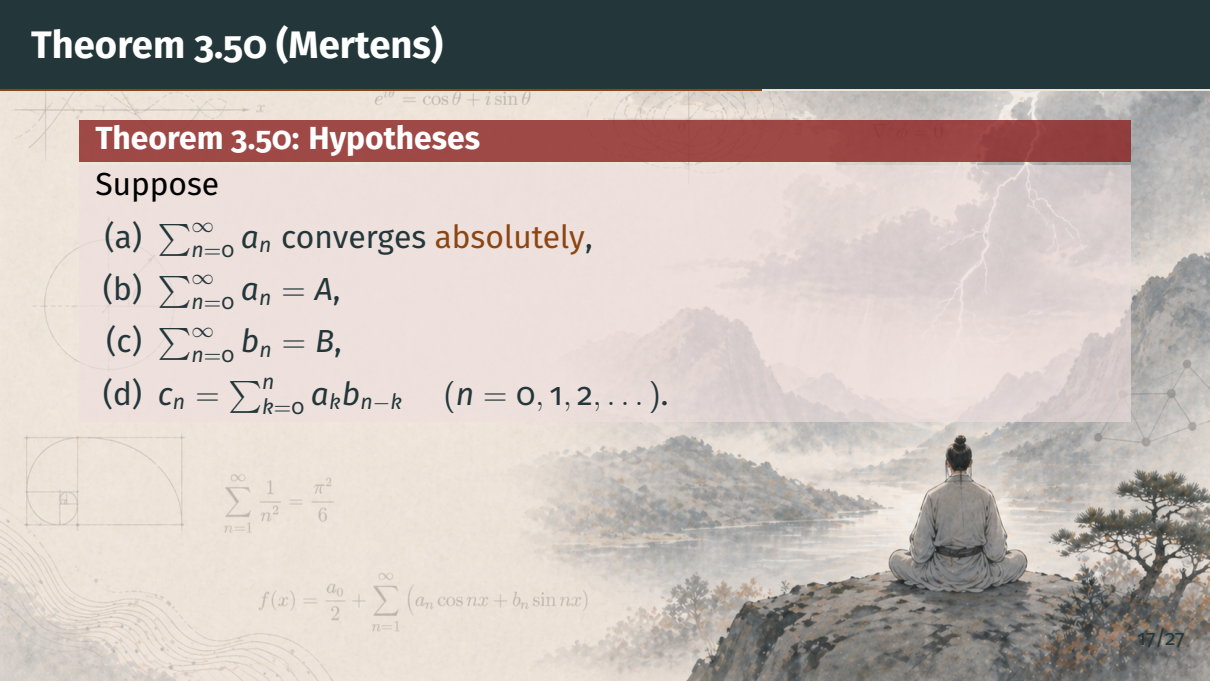


Theorem 3.50 (Mertens)

Theorem 3.50: Hypotheses

Suppose

- (a) $\sum_{n=0}^{\infty} a_n$ converges **absolutely**,
- (b) $\sum_{n=0}^{\infty} a_n = A$,
- (c) $\sum_{n=0}^{\infty} b_n = B$,
- (d) $c_n = \sum_{k=0}^n a_k b_{n-k} \quad (n = 0, 1, 2, \dots).$


$$\sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{\pi^2}{6}$$

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} (a_n \cos nx + b_n \sin nx)$$

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- (d) $c_n = \sum_{k=0}^n a_k b_{n-k} \quad (n = 0, 1, 2, \dots).$

Conclusion

$$\sum_{n=0}^{\infty} c_n = AB$$

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} (a_n \cos nx + b_n \sin nx)$$

Proof Strategy: Theorem 3.50

Goal: show $C_n = \sum_{k=0}^n c_k \rightarrow AB$.

Approach:

1. Regroup C_n as a combination of the a_k and the partial sums B_k
2. Split off the **main term** $A_n B$ (which $\rightarrow AB$) plus a remainder γ_n
3. Show $\gamma_n \rightarrow 0$: split it at a cutoff N where B_k is close to B
4. Control the tail using $\alpha = \sum |a_n|$ — **this is where absolute convergence enters**

Proof of Theorem 3.50 — Step 1: Regrouping C_n

Step 1: Express C_n via the partial sums B_k

Put $A_n = \sum_{k=0}^n a_k$, $B_n = \sum_{k=0}^n b_k$, $C_n = \sum_{k=0}^n c_k$. Then

$$\begin{aligned} C_n &= a_0 b_0 + (a_0 b_1 + a_1 b_0) + \cdots + (a_0 b_n + a_1 b_{n-1} + \cdots + a_n b_0) \\ &= a_0 B_n + a_1 B_{n-1} + \cdots + a_n B_0 \end{aligned}$$

Technique: regroup by the **a-factor** instead of by diagonal.

$$\sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{\pi^2}{6}$$

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} (a_n \cos nx + b_n \sin nx)$$

Proof of Theorem 3.50 — Step 2: Main Term + Remainder

Step 2: Split off $A_n B$

Put $\beta_n = B_n - B$ (so $\beta_n \rightarrow 0$). Then

$$\begin{aligned} C_n &= a_0(B + \beta_n) + a_1(B + \beta_{n-1}) + \cdots + a_n(B + \beta_0) \\ &= A_n B + \underbrace{a_0\beta_n + a_1\beta_{n-1} + \cdots + a_n\beta_0}_{= \gamma_n} \end{aligned}$$

Since $A_n B \rightarrow AB$, it suffices to show

$$\sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{\pi^2}{6}$$

$$\lim_{n \rightarrow \infty} \gamma_n = 0.$$

(21)

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} (a_n \cos nx + b_n \sin nx)$$

Proof of Theorem 3.50 — Step 3: Splitting γ_n

Step 3: Cut at a threshold N

Put $\alpha = \sum_{n=0}^{\infty} |a_n|$ [here we use (a)!]

Given $\varepsilon > 0$: since $\beta_n \rightarrow 0$, choose N with $|\beta_n| \leq \varepsilon$ for $n \geq N$. Then

$$\begin{aligned} |\gamma_n| &\leq |\beta_0 a_n + \cdots + \beta_N a_{n-N}| + |\beta_{N+1} a_{n-N-1} + \cdots + \beta_n a_0| \\ &\leq |\beta_0 a_n + \cdots + \beta_N a_{n-N}| + \varepsilon \alpha \end{aligned}$$

$$\sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{\pi^2}{6}$$

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} (a_n \cos nx + b_n \sin nx)$$

Proof of Theorem 3.50 — Step 4: Letting $n \rightarrow \infty$

Step 4: The first piece dies out

Keep N **fixed**, let $n \rightarrow \infty$. Since $a_k \rightarrow 0$ as $k \rightarrow \infty$,

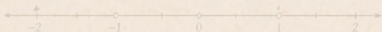
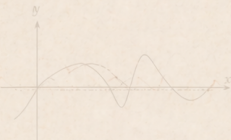
$$|\beta_0 a_n + \cdots + \beta_N a_{n-N}| \rightarrow 0 \implies \limsup_{n \rightarrow \infty} |\gamma_n| \leq \varepsilon \alpha.$$

Since ε is arbitrary, (21) follows. ■


$$\sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{\pi^2}{6}$$

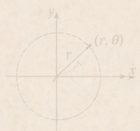
$$\sum_{n=0}^{\infty} c_n = AB$$

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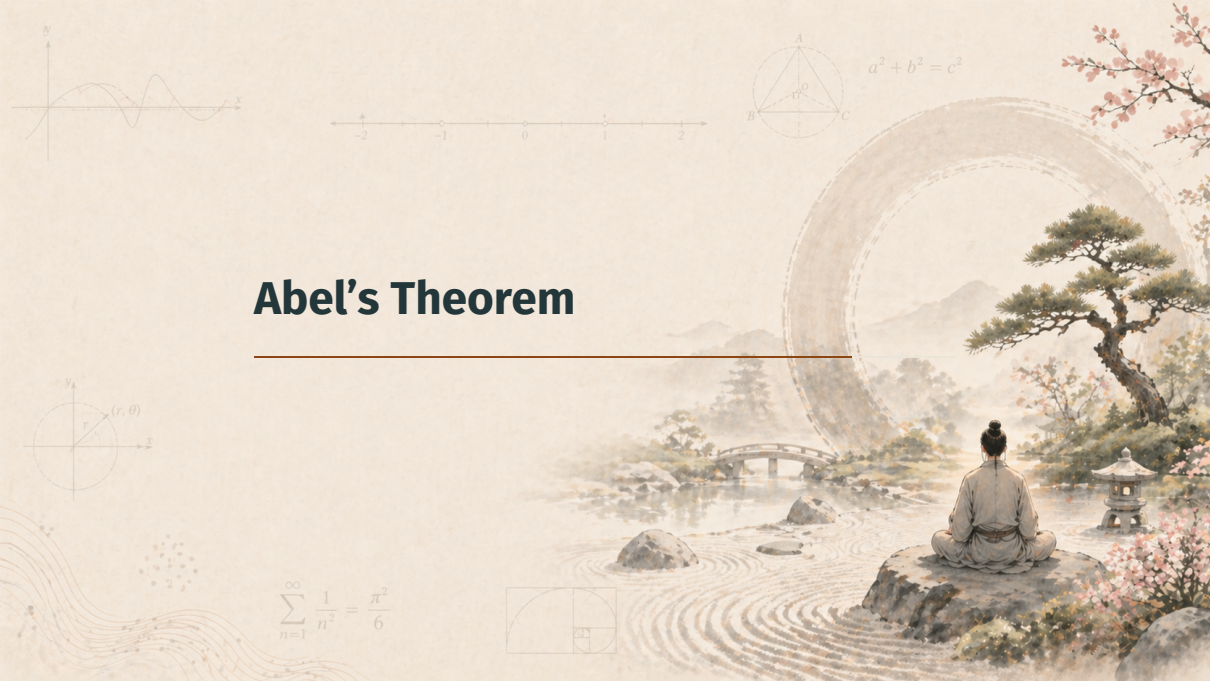


$$a^2 + b^2 = c^2$$

Abel's Theorem



$$\sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{\pi^2}{6}$$



Theorem 3.51 (Abel)

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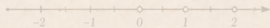
If the series $\sum a_n$, $\sum b_n$, $\sum c_n$ converge to A , B , C , and $c_n = a_0 b_n + \cdots + a_n b_0$, then

$$C = AB.$$

- **No** absolute convergence is assumed!
- If the Cauchy product converges, it converges to the **right value**
- Proof deferred: it uses the continuity of power series (Chapter 8)

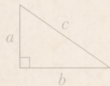
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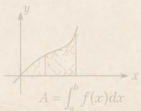


$$x^2 + y^2 = r^2$$

Summary



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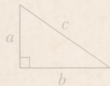
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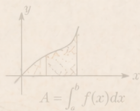
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- **Addition (3.47):** convergent series add term by term:
 $\Sigma(a_n + b_n) = A + B, \Sigma ca_n = cA$



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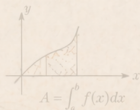
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- **Danger (3.49):** the product of two **nonabsolutely** convergent series can **diverge**



$$a^2 + b^2 = c^2$$



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- **Danger (3.49):** the product of two **nonabsolutely** convergent series can **diverge**
- **Mertens (3.50):** one **absolutely** convergent factor $\Rightarrow \Sigma c_n = AB$;
Abel (3.51): if Σc_n converges, its sum is AB

Coming up next: Rearrangements — can reordering change the sum?